Cervical Shortening 05. Dilation, Rescue Cerclage, and Periviability

Companion reading handout for simultaneous listening.
Audio course on cervical shortening during pregnancy. Episode five. Dilation, rescue cerclage, and periviability.

Opening frame

This episode begins with a warning.

Do not treat cervical dilation as a more dramatic version of short cervical length.

It is different.

A closed cervix measuring eighteen millimeters on TV ultrasound is a risk marker. Painless cervical dilation with visible membranes at twenty one weeks is an acute high-risk presentation. The patient may still be asymptomatic. She may not be contracting. But the anatomy, infection risk, membrane risk, and counseling problem have changed.

Dominant model

The dominant model is this.

Dilation moves the resident from prediction to triage.

Now the first question is not, should I give progesterone? The first question is, what is happening right now, and is pregnancy prolongation safe?

Definitions and categories

Let us define the state.

Physical-exam-indicated cervical insufficiency usually means painless cervical dilation in the current pregnancy, often with visible or prolapsing membranes, before viability or in the periviable period, in the absence of another explanation such as labor, PPROM, infection, bleeding, or major fetal anomaly.

That definition is not just academic. It tells you what must be excluded.

Before considering exam-indicated cerclage, assess contractions. If the patient is in active labor, cerclage is not a treatment for labor.

Assess membranes. If there is PPROM, placing a cerclage may increase infectious and maternal-fetal risk.

Assess infection. Fever, uterine tenderness, purulent discharge, maternal tachycardia, significant leukocytosis, or other local signs of intraamniotic infection change the goal. Pregnancy prolongation in infection can be dangerous.

Assess bleeding and abruption concern. Significant bleeding may represent a different trigger for preterm birth and may make cerclage unsafe.

Assess fetal status and anatomy. Fetal demise or major anomaly changes counseling and goals.

This is the first resident reflex of the episode:

Before rescue cerclage, prove that you are not sewing through labor, ruptured membranes, infection, bleeding, or a changed fetal goal of care.

Now let us talk about why infection matters so much.

Midtrimester dilation can be associated with occult intraamniotic infection. Gabbe emphasizes that positive amniotic fluid cultures are common in women with midtrimester cervical dilation. This is why some centers consider amniocentesis in selected cases before rescue cerclage, especially when membranes are exposed or inflammatory concern exists. Local practice varies. The principle does not vary: infection changes risk and prognosis.

Now the guideline frame.

NICE recommends considering emergency cerclage for women between sixteen plus zero and twenty seven plus six weeks when the cervix is dilated and membranes are exposed but unruptured. NICE recommends against emergency cerclage when there are signs of infection, active vaginal bleeding, or uterine contractions.

FIGO similarly frames rescue cerclage as individualized. The potential benefit is pregnancy prolongation. The risk is procedure-related rupture, infection, bleeding, or simply performing an intervention in a pregnancy already moving through the common pathway.

This is not a checkbox decision. It is a risk conversation.

Now compare with ultrasound-indicated cerclage.

Ultrasound-indicated cerclage asks: in a high-risk patient with a short but closed cervix, can a stitch reduce early birth?

Exam-indicated cerclage asks: in a patient whose cervix is already open, can we safely reverse or slow a process that may already include membrane exposure and inflammation?

Different question. Different risk. Different counseling.

Let us build the bedside sequence.

A patient presents at twenty two weeks after an ultrasound shows no measurable cervix. Speculum exam shows membranes visible at the external os. She reports pelvic pressure but no clear contractions. No bleeding. No leakage of fluid.

Step one: stabilize the room. This is not a routine clinic follow-up anymore.

Step two: confirm gestational age, dating, fetal status, and anatomy information.

Step three: assess symptoms. Contractions, pain, pressure, bleeding, leakage, fever, discharge.

Step four: assess membranes. Use the local PPROM diagnostic protocol. Do not assume intact membranes because the patient is not leaking heavily.

Step five: assess infection and bleeding.

Step six: involve the right level of care. This is MFM level decision-making, and if the gestational age is near viability, neonatal counseling and transfer capability may matter.

Step seven: discuss whether exam-indicated cerclage is appropriate.

Now the patient asks, what are the chances?

Do not give false precision. Prognosis depends on gestational age, degree of dilation, membrane exposure or prolapse, infection status, prior history, cervical tissue, and whether contractions begin. A patient one centimeter dilated with membranes just visible is not the same as a patient five centimeters dilated with membranes prolapsing into the vagina.

A useful counseling line is:

This is different from a short cervix on ultrasound. The cervix is already open, and the membranes are exposed. Sometimes a cerclage can prolong pregnancy in carefully selected patients, but only if there is no labor, no ruptured membranes, no infection, and no significant bleeding. The procedure itself can also rupture the membranes or introduce risk. We need to decide whether a stitch is more likely to help than harm in your exact situation.

Now periviability.

Once cervical shortening or dilation creates a credible threat of birth near the limit of viability, the lecture expands beyond the cervix.

The resident must think about neonatal prognosis, parental goals, steroids, magnesium, transfer, and what interventions are appropriate at that gestational age.

This is where timing becomes ethically and clinically dense.

At very early gestational ages, the question may be whether neonatal resuscitation would be offered or desired. As viability approaches, the question may shift toward antenatal corticosteroids, magnesium sulfate for neuroprotection when early preterm delivery is expected and gestational age fits local protocol, tocolysis only if it creates a useful steroid or transfer window, and delivery planning if maternal or fetal status changes.

Do not confuse prevention with acute preterm birth management.

Progesterone and cerclage are prevention or prolongation strategies in selected phenotypes.

Antenatal corticosteroids are fetal preparation when preterm birth risk is sufficiently high.

Magnesium is neuroprotection when early preterm delivery is anticipated.

Tocolysis is a short-term tool, not long-term treatment of the cervix.

GBS prophylaxis belongs to threatened preterm birth management according to protocol.

Antibiotics are not for preterm labor with intact membranes unless there is another indication. PPROM is different.

Now let us discuss cerclage and PPROM.

If PPROM occurs with a cerclage in place, Gabbe notes that retention has not been shown to improve neonatal outcomes and early cerclage removal is recommended when PROM occurs. Some clinicians consider very short-term retention during steroid administration in selected cases, but this is uncertain and must be weighed against infection risk.

The teaching point is that cerclage does not make PPROM a simple problem. It may complicate infection and management decisions.

Now a case.

Clinical cases

Twenty one weeks, singleton, no contractions, painless dilation two centimeters, membranes visible, afebrile, no bleeding, membranes appear intact.

This is a possible exam-indicated cerclage case. But the resident cannot jump straight to the OR. The workup must exclude infection, labor, PPROM, and bleeding. Counseling must include procedural risks and uncertain prognosis.

Second case.

Twenty two weeks, cervix three centimeters, membranes visible, painful contractions every three minutes.

This is not the same problem. Active contractions suggest labor. Cerclage is unlikely to stop established labor and may be harmful.

Third case.

Twenty three weeks, visible membranes, maternal fever, uterine tenderness, and purulent discharge.

Do not rescue-cerclage infection. The priority is maternal safety and infection management.

Fourth case.

Twenty four weeks, advanced dilation, intact membranes, no infection, no contractions. Neonatal resuscitation is desired if delivery occurs.

Now the plan must include both the cerclage decision and periviability preparation. Neonatal counseling, steroid timing, magnesium if delivery becomes imminent, and transfer to an appropriate center may all be part of the management.

Final synthesis

Let us close with three resident reflexes.

First: dilation is not just a shorter cervix. It is a new clinical category.

Second: before exam-indicated cerclage, exclude labor, PPROM, infection, bleeding, and fetal conditions that change goals of care.

Third: near viability, do not let the cervix monopolize your thinking. The real job is integrated maternal-fetal-neonatal planning.

The cervix may be the doorway into the case, but once it opens, the entire periviability system may enter the room.